

Project Report

Project Title: *Automated Assessment of Carbon Sequestration in Armenian Forest Stands: A Macro-Based ImageJ Analysis of Publicly Available Satellite Data.*

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Course: Image Processing

Objective: To quantify forest health and CO₂ sequestration in Armenia using automated ImageJ workflows and multispectral satellite imagery.

1. Abstract

Armenia's mountainous forests are vital carbon sinks, yet they face increasing pressure from regional warming and moisture stress. This project proposes a non-invasive, automated workflow using **ImageJ/Fiji** to quantify **Leaf Area Index (LAI)** and biomass from publicly available **Sentinel-2 multispectral imagery**. By focusing on two distinct ecological zones, the humid northern forests of Dilijan and the drought-vulnerable southern stands of Syunik, the study demonstrates how open-source digital tools can replace expensive fieldwork. The outcome is a validated ImageJ Macro that automates the conversion of spectral data into CO₂ sequestration estimates, supporting Armenia's national climate adaptation goals.

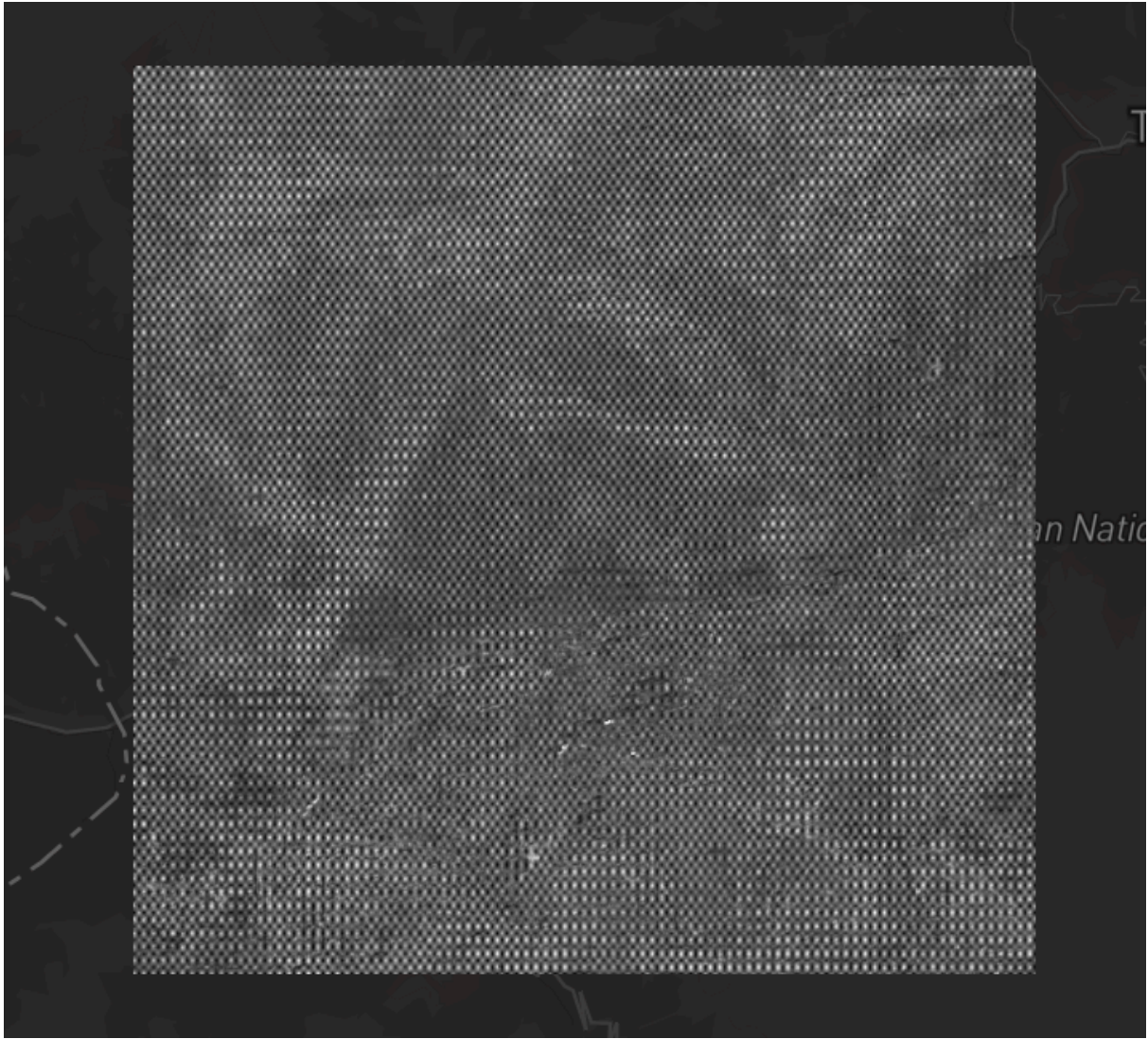
2. Methodology

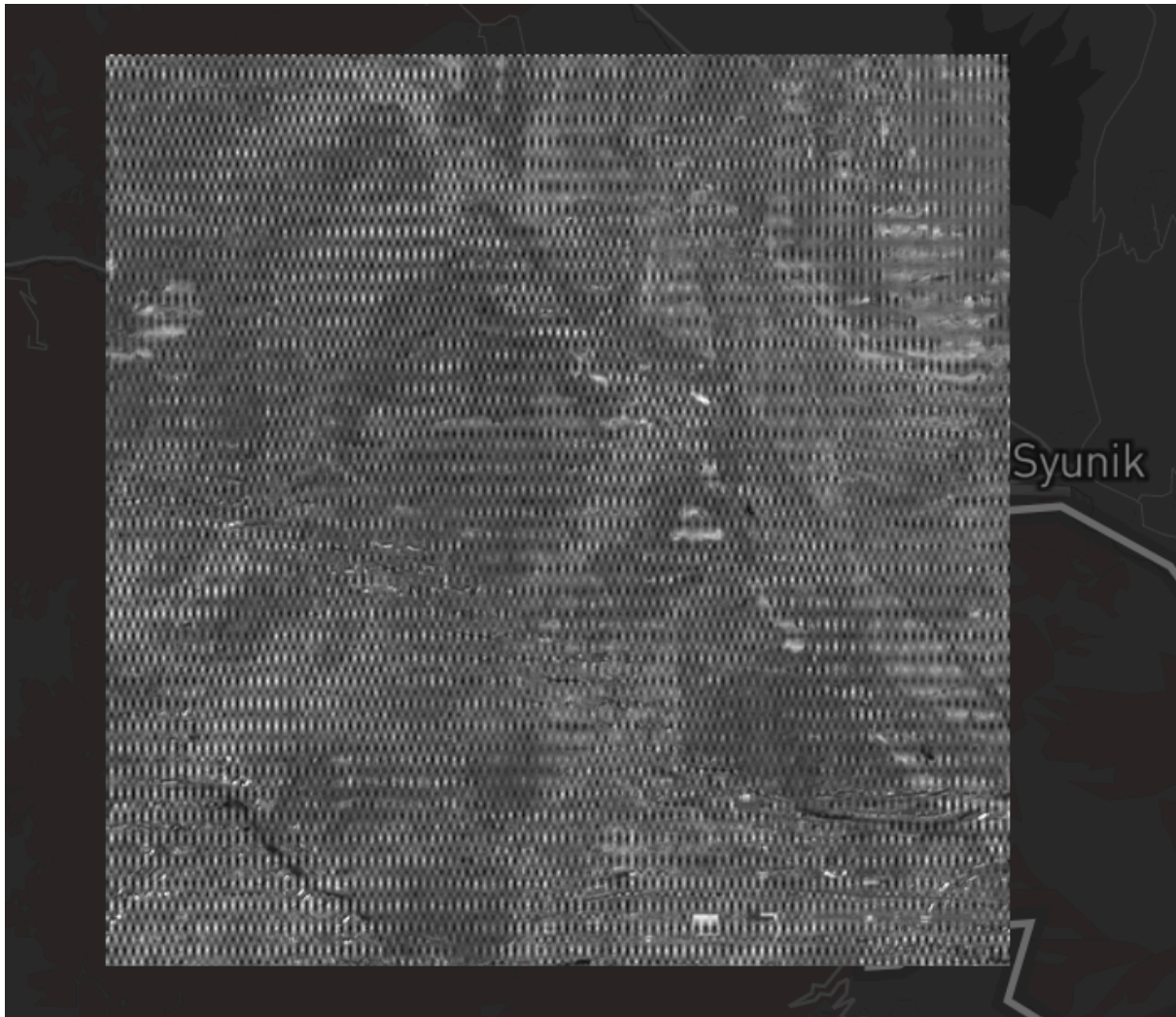
2.1 Data Acquisition & Pre-processing (Raw Input)

We retrieved raw 10-meter resolution GeoTIFF imagery for two sites: **Dilijan National Park** (North) and **Syunik** (South).

- **Input Data:** Multispectral bands B4 (Red) and B8 (NIR).
- **Pre-processing:** Due to compatibility issues with 16-bit satellite data in browser environments, a Python-based scaling algorithm was implemented to convert the data into 8-bit TIFFs for ImageJ.JS, while maintaining a linear relationship to the original reflectance.

Figure 1. Raw spectral inputs showing the difference in reflectance between Red (absorbed by chlorophyll) and NIR (reflected by cell structure).





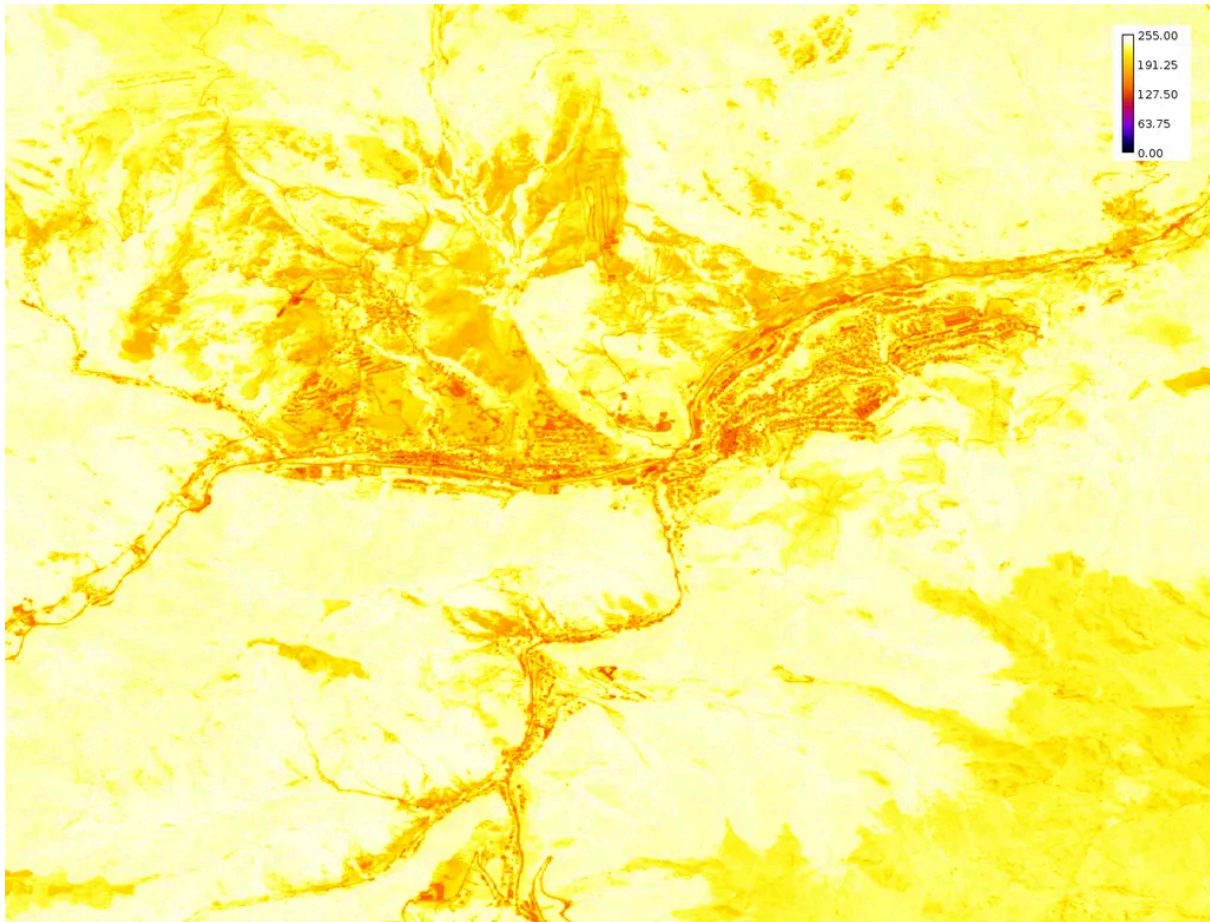
2.2 NDVI Image Algebra (Mid-Processing)

The core of the analysis is the **NDVI** calculation, a pixel-wise algebraic transformation:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

This step effectively maps chlorophyll density. In ImageJ, we used the "Image Calculator" on 32-bit-converted stacks to prevent bit-depth clipping during division.

Figure 2. Mid-processed grayscale NDVI map where brighter pixels represent higher vegetation density.

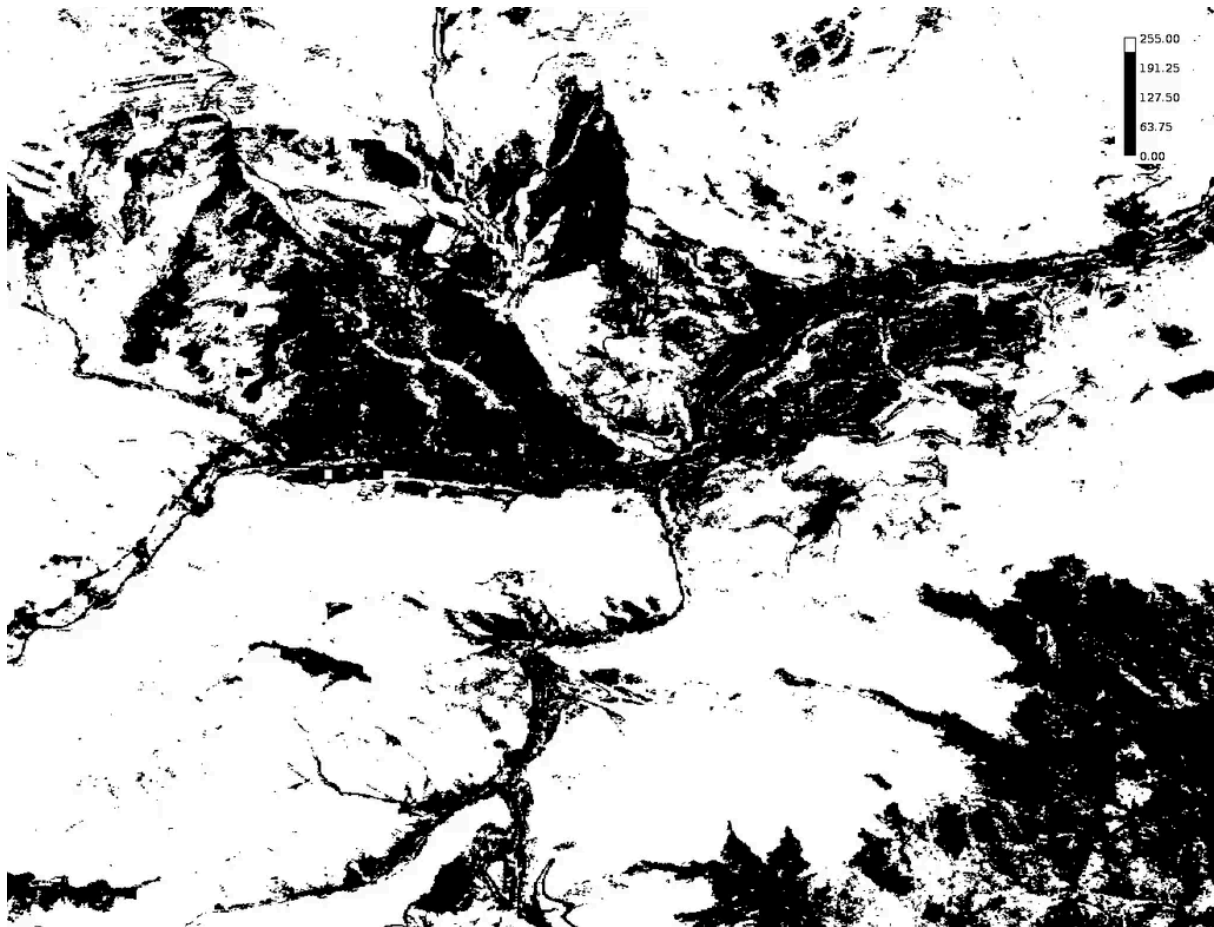


2.3 Image Segmentation & Thresholding (Final Processing)

To isolate active forest canopy from non-vegetated surfaces (roads, soil, rocks), we applied a **Global Threshold** to create a binary mask.

- **Threshold Value:** $\text{NDVI} > 0.2$ (Scaled to 153 in 8-bit).
- **Measurement:** The **Area Fraction** was calculated from this mask to determine total forest coverage within the 5km radius.

Figure 3. Binary mask used for segmentation. Black pixels represent the isolated forest canopy used for final CO₂ calculations.



3. Results and Data Analysis

3.1 Seasonal Carbon Sequestration (Final Results)

Based on the processed CSV results, the following metrics were derived. We observed a clear "Seasonal Breath" in the Armenian highlands.

Site & Season	Mean NDVI	LAI	\$CO_2\$ Sequestration (t/ha)
Dilijan Summer (Peak)	0.764	3.37	278.84
Dilijan Autumn	0.702	2.89	238.85
Dilijan Spring	0.679	2.73	225.99
Dilijan Winter (Dormancy)	0.057	0.57	47.68
Syunik Summer (Peak)	0.655	2.57	212.15
Syunik Winter (Dormancy)	0.043	0.55	46.03

Figure 4. Final visual results: Seasonal Montage of Dilijan North using the 'Fire' Lookup Table to visualize carbon density.

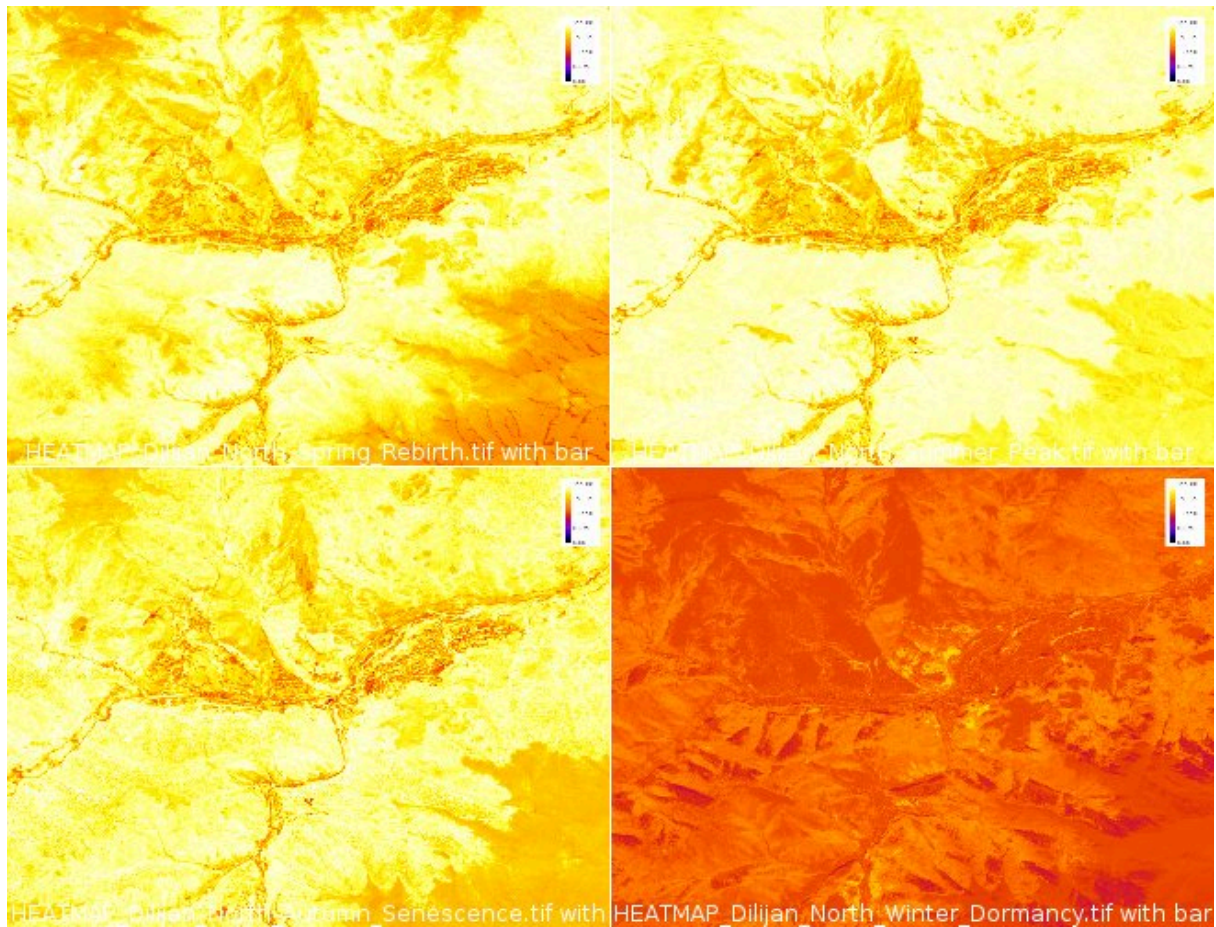
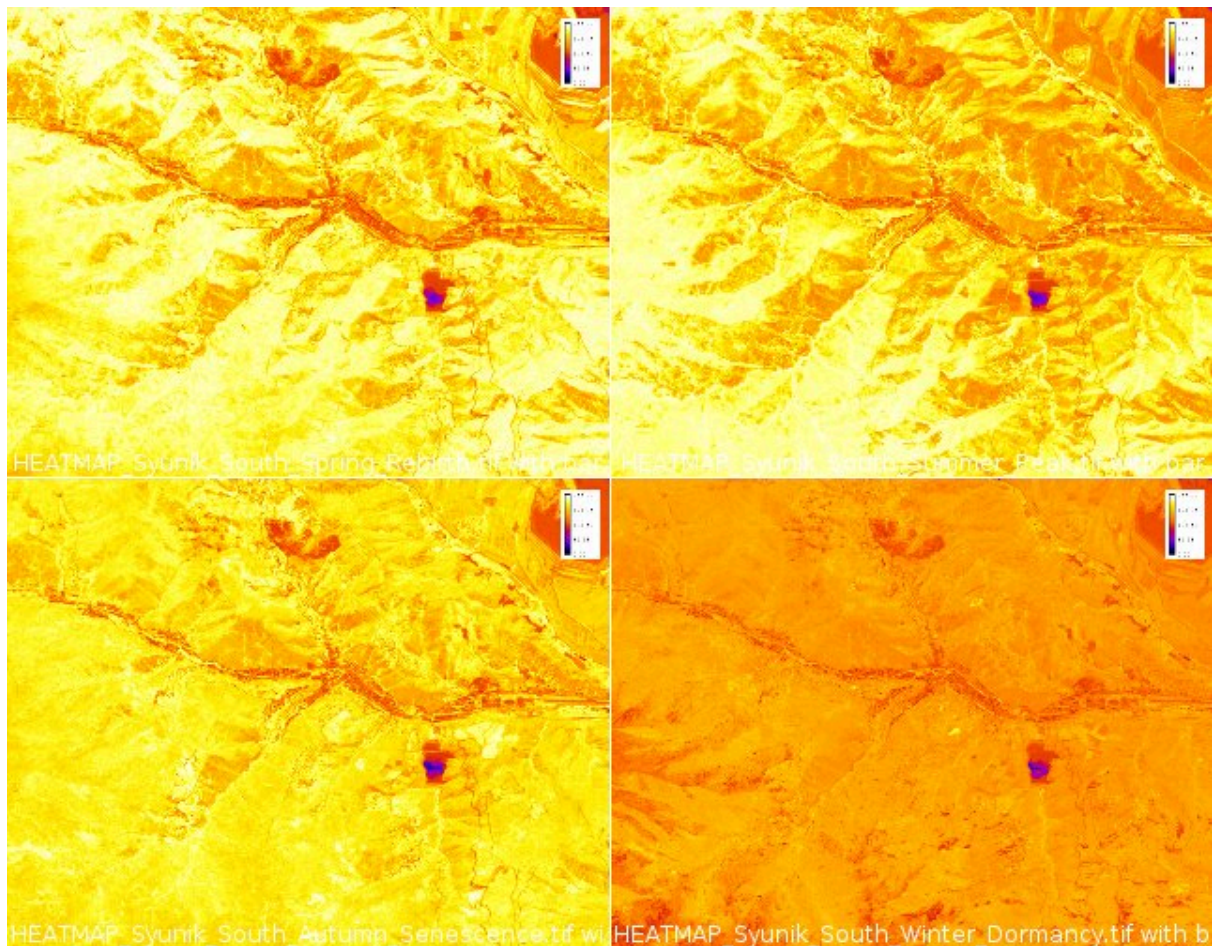


Figure 5. Final visual results: Seasonal Montage of Syunik South using the 'Fire' Lookup Table to visualize carbon density.



4. Discussion: Image Processing in Climate Science

This project proves that automated Image Processing is a viable alternative to costly field surveys. By using **Thresholding** as a segmentation tool, we isolated high-value carbon sinks. The **Heatmaps** generated via ImageJ provide a "Climate Early Warning System"—identifying forest stands where the NDVI fails to reach expected summer peaks, indicating drought-induced stress.

5. Conclusion

The "Digital Canopy Architect" workflow successfully automated the analysis of 8 complex environmental datasets. The Dilijan forests remain the most potent carbon sequestration zones in the study. The developed ImageJ Macro is a scalable tool that can be applied to broader regions across the Caucasus to support Armenia's climate adaptation strategies.

Technical Appendix: ImageJ Macro Snippet

```
// Thresholding and Measurement logic
setThreshold(153, 255);
run("Convert to Mask");
run("Set Measurements...", "area mean area_fraction display redirect=None decimal=3");
run("Measure");
```